

# IE598 Machine Learning in Finance Lab

## Module 5 Homework (Dimensionality Reduction)

Rong Gan (ronggan2)

### Introduction

This homework analyzes the Treasury Yield Curve dataset using dimensionality reduction and regression models. The dataset contains 30 predictor attributes and one target variable, `Adj_Close`. First, we perform exploratory data analysis to examine the structure of the dataset and the relationships among the variables. Principal Component Analysis (PCA) is then used to reduce the 30 predictor attributes to three principal components. Finally, Linear Regression and Support Vector Regression (SVR) are fitted using both the original 30 attributes and the three principal components. The models are evaluated using  $R^2$ , RMSE, and training time.

## 1 Exploratory Data Analysis

### 1.1 Dataset Description

The dataset used in this analysis is the Treasury Yield Curve dataset. It contains 8,071 daily observations from January 13, 1987 to May 17, 2019, with 31 columns in total. The dataset includes 30 numerical predictor attributes, `SVENF01–SVENF30`, which represent the yield on Treasury bond instruments with maturities ranging from 1 year to 30 years. The target variable, `Adj_Close`, is the daily adjusted closing price of the PIMCO Total Return Bond Fund (Ticker: `PTTAX`).

### 1.2 Data Preparation

The target variable is `Adj_Close`, while the remaining 30 variables are used as predictors. The dataset is divided into training and test sets using an 85–15 split with `random_state=42`. This same split is used for all subsequent models so that their performance can be compared under the same test observations. The training set contains 6,860 observations, while the test set contains 1,211 observations.

### 1.3 Data Visualizations

The distribution of the target variable, `Adj_Close`, is shown in Figure 1. We see that the values are widely dispersed across the observed range, with a slightly higher concentration at lower values. Figure 2 shows the movement of `Adj_Close` over time. The target variable generally increases over the observed period, although there are fluctuations throughout the series.

The distributions of the 30 predictor attributes are shown in Figure 3. We observe that some of the predictor attributes have similar distributional shapes, suggesting that they may contain related information. To further examine the relationships among the predictors, the correlation structure of the 30 predictor attributes is shown in Figure 4. The heatmap shows that the predictor attributes are highly correlated with one another, indicating that the original 30-dimensional feature space may contain redundant information.

The strong correlations among the predictors provide motivation for using PCA. PCA can combine correlated variables into a smaller number of orthogonal principal components while retaining most of the variation in the

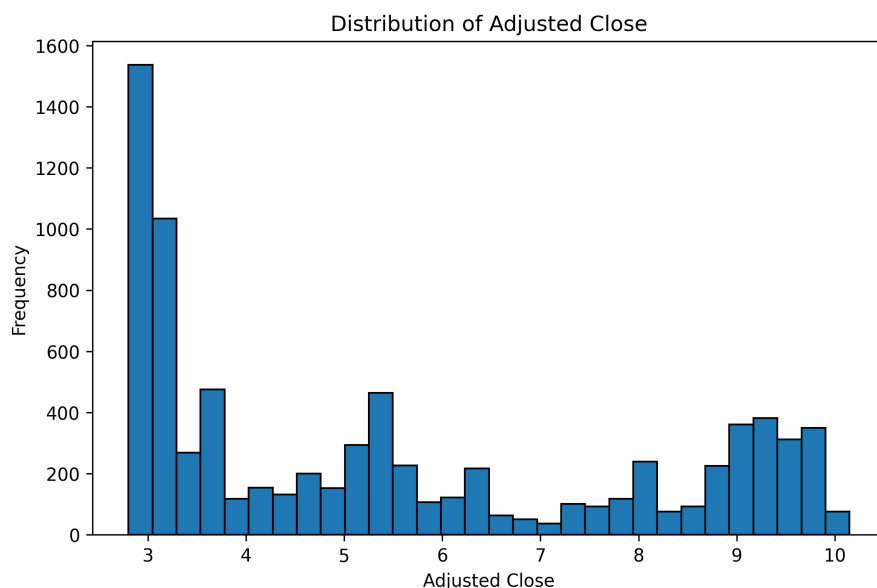


Figure 1: Distribution of Adj\_Close

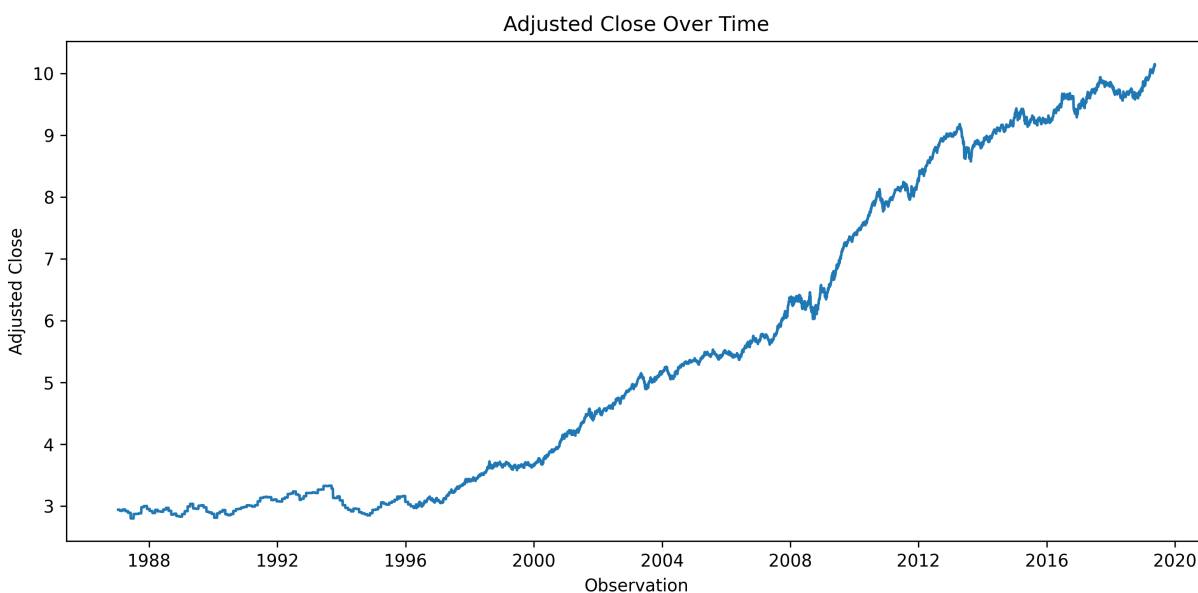


Figure 2: Adj\_Close Over Time

original data.

## 2 Principal Component Analysis

### 2.1 PCA with All Components

Before applying PCA, the 30 predictor attributes are standardized using the training data. The standardization ensures that the PCA is not dominated by variables with larger numerical scales.

PCA is first fitted using all 30 components to examine the proportion of variance explained by each principal

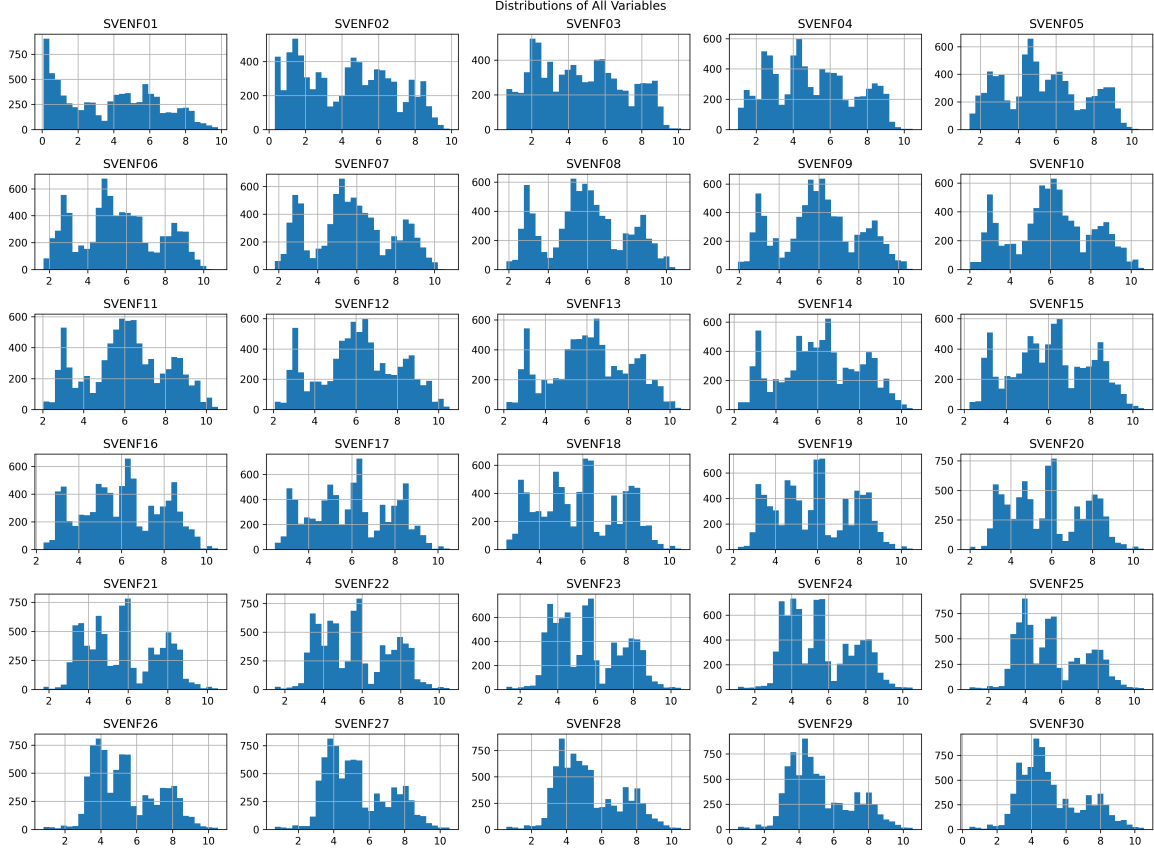


Figure 3: Distributions of the 30 Predictor Attributes

component. Figure 5 shows the explained variance ratio for the principal components. The explained variance decreases rapidly after the first component. This indicates that a small number of principal components may represent most of the variation contained in the original 30 attributes.

## 2.2 PCA with Three Components

We again perform PCA and retain only three principal components. Figure 6 shows the cumulative explained variance of the first three principal components. The three principal components reduce the predictor space from 30 attributes to three components while retaining 99.44% of the total variance in the original predictors.

## 3 Regression Models

Two regression methods are considered: Linear Regression and SVR. Each method is fitted using both the original 30 standardized predictor attributes and the three principal components obtained from PCA, resulting in four models for comparison. Linear Regression provides a linear baseline, while SVR can capture nonlinear relationships between the predictors and the target variable. Comparing the baseline and PCA versions of each model allows us to examine whether reducing the predictor space from 30 attributes to three principal components affects predictive performance.

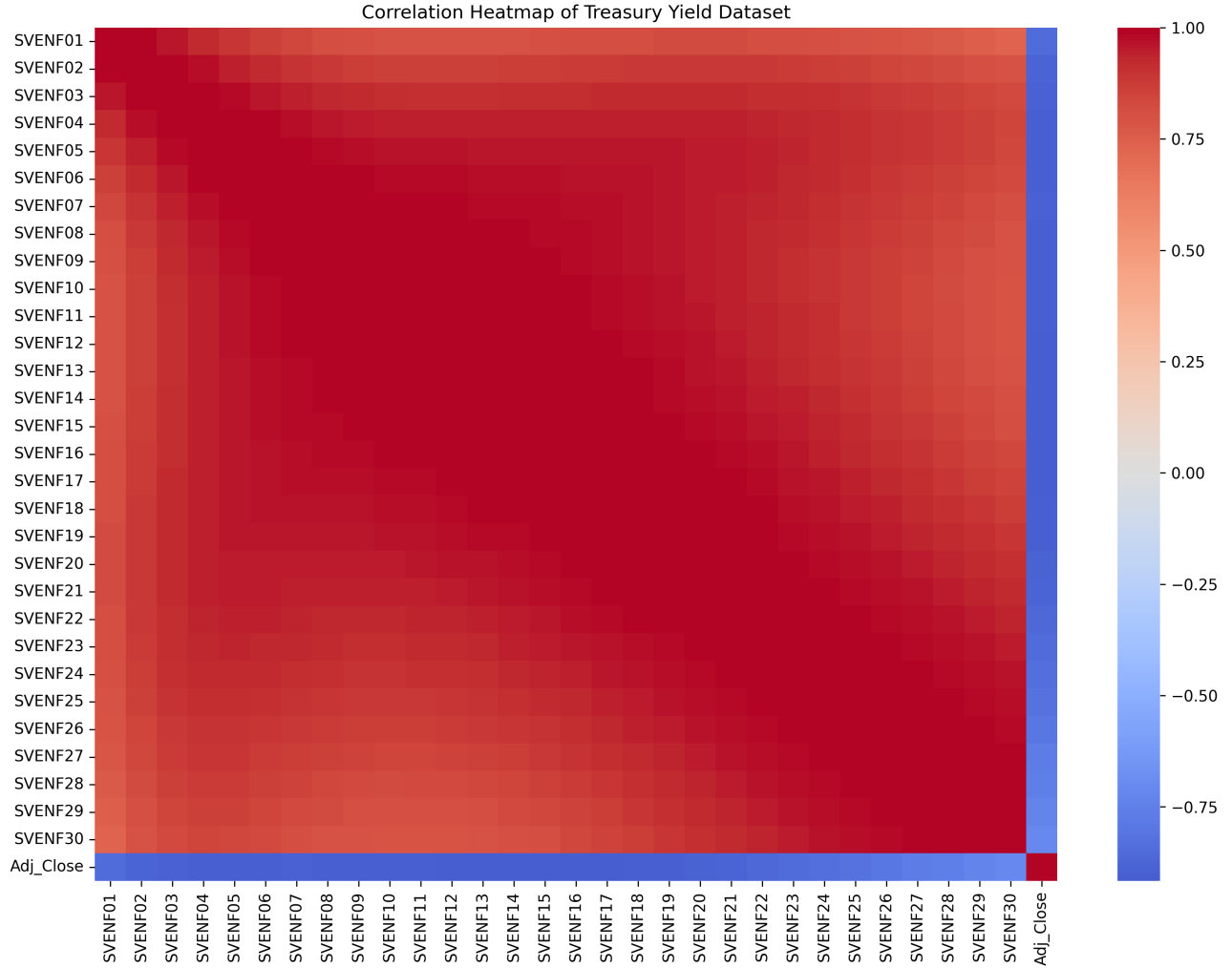


Figure 4: Correlation Heatmap of the 30 Predictor Attributes

## 4 Model Comparison

The performance of models is evaluated using  $R^2$  and RMSE. A higher  $R^2$  indicates that the model explains more of the variation in the target variable, while a lower RMSE indicates smaller prediction errors. The performance of the four regression models is summarized in Table 1. The comparison includes both training and test performance, as well as training time.

Table 1: Performance Comparison of Regression Models

|                     | Linear Baseline | SVR Baseline | Linear PCA | SVR PCA  |
|---------------------|-----------------|--------------|------------|----------|
| Train $R^2$         | 0.902273        | 0.988651     | 0.867218   | 0.977966 |
| Test $R^2$          | 0.904131        | 0.989128     | 0.866242   | 0.977669 |
| Train RMSE          | 0.776653        | 0.264669     | 0.905294   | 0.368779 |
| Test RMSE           | 0.782370        | 0.263472     | 0.924131   | 0.377595 |
| Training Time (sec) | 0.030079        | 6.308690     | 0.005784   | 2.762950 |

Using the original 30 predictor attributes, both regression methods achieve strong  $R^2$ , while Support Vector Regression producing smaller prediction errors than Linear Regression. After applying PCA, the predictive

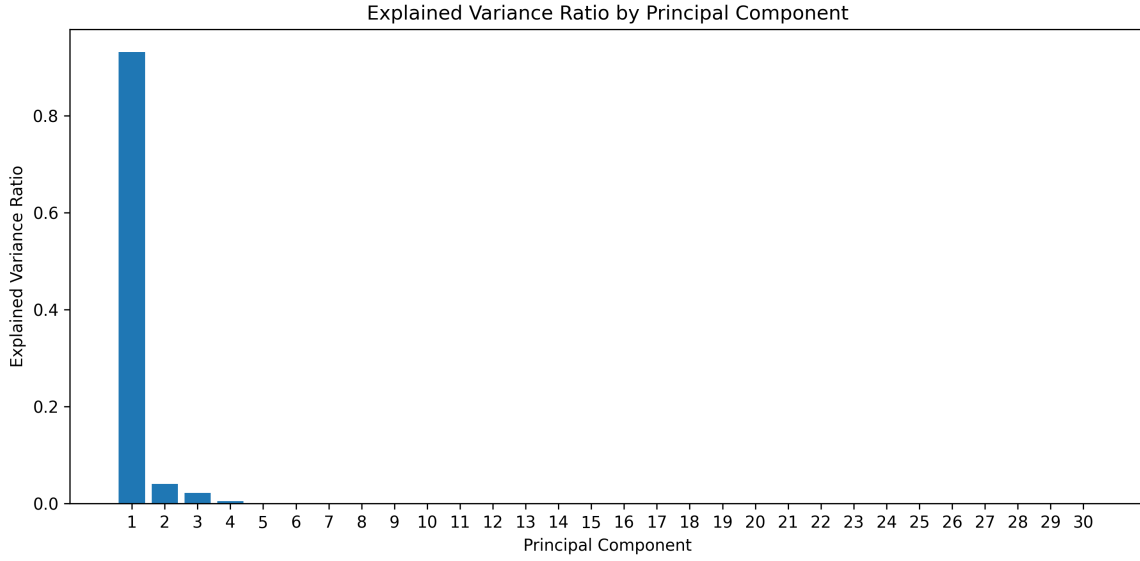


Figure 5: Explained Variance Ratio of Principal Components

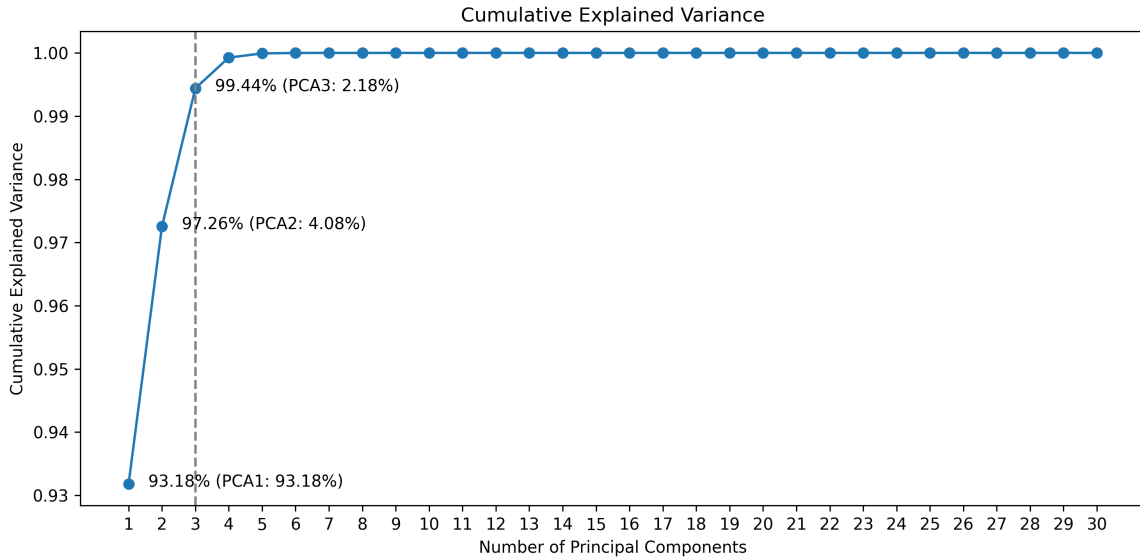


Figure 6: Cumulative Explained Variance of the First Three Principal Components

performance of both models decreases slightly. For Linear Regression, the test  $R^2$  changes from 0.904 to 0.866 and the test RMSE changes from 0.782 to 0.924. For Support Vector Regression, the test  $R^2$  changes from 0.989 to 0.978 and the test RMSE changes from 0.263 to 0.378. These changes indicate that PCA results in a small sacrifice in predictive performance.

We also provide the actual versus predicted values for the four models in Figure 7 for further comparison of their prediction performance. The x-axis represents the actual `Adj_Close` values, while the y-axis represents the predicted values. Each point corresponds to one observation in the test data, and the dashed 45-degree line represents perfect prediction, where the predicted value equals the actual value. We observe that the SVR predictions are more closely concentrated around the 45-degree line, indicating smaller prediction errors compared with Linear Regression. The differences between the baseline and PCA versions of both models are not very apparent from the plot, the PCA predictions appear to be slightly more dispersed from the 45-degree line. Moreover, PCA also reduces the training time for both models. The training time decreases from 0.03 to 0.006

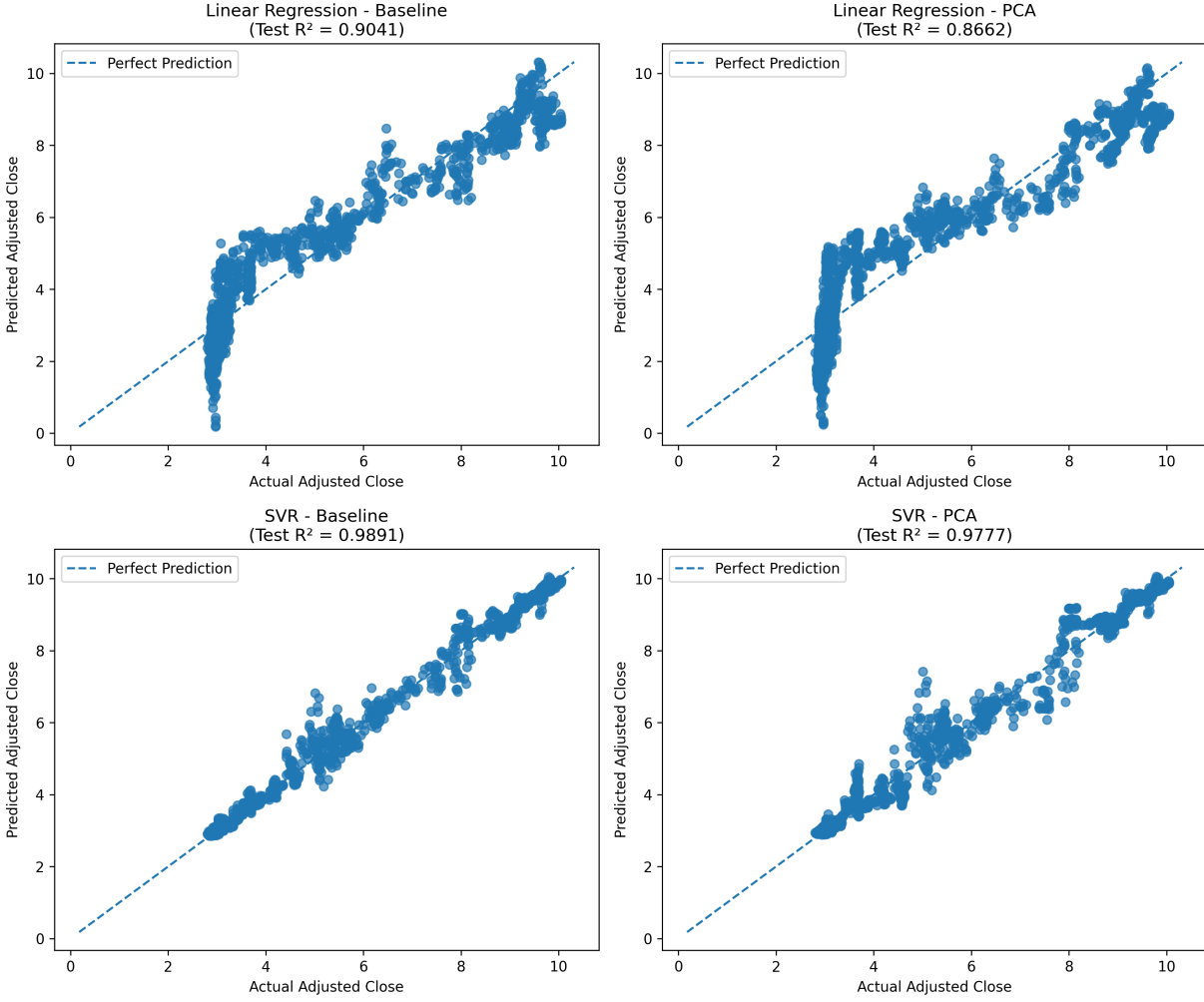


Figure 7: Actual vs. Predicted Adj\_Close for the Four Regression Models

seconds for Linear Regression and from 6.309 to 2.763 seconds for Support Vector Regression. Therefore, in this experiment, PCA provides a trade-off between predictive performance and computational efficiency: a substantial reduction in the number of predictor attributes and training time is accompanied by relatively small loss in predictive performance.

## 5 Conclusions

This analysis compared Linear Regression and Support Vector Regression using both the original 30 predictor attributes and a three-component PCA representation. The predictor attributes are highly correlated, making dimensionality reduction appropriate for this dataset. PCA reduces the original 30-dimensional feature space to three principal components while retaining 99.44% of the total variance.

On the untransformed data, Support Vector Regression achieves the best predictive performance. PCA does not improve predictive performance for either model; instead, it leads to a slight decrease in performance. However, PCA reduces training time for both Linear Regression and Support Vector Regression, providing a trade-off between predictive performance and computational efficiency.

## A GitHub Repository

The code and analysis for this report are available in the GitHub repository:

<https://github.com/ronggan01/Machine-Learning-in-Finance-Fall-2026>